

The Sunshine Equation: Exploring Melanoma in Spain through Spatial Analysis

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Abstract

Spain has one of the highest skin cancer rates in Europe, with a rising incidence of melanoma, actinic keratosis, and squamous cell carcinoma, primarily attributed to prolonged sun exposure in its Mediterranean climate. While previous studies have analyzed melanoma mortality across provinces, this study shifts the focus to the incidence of melanoma from a spatial perspective. I investigate the potential influence of factors such as immigration, average age, proximity to hospitals and healthcare facilities, UV index, and income on melanoma distribution across Spain. To achieve this, I will use R for data collection, cleaning, and storage, while ArcGIS will be employed for spatial analysis. By conducting an in-depth spatial study, this research aims to identify significant patterns and correlations that may contribute to a better understanding of melanoma incidence in the country.

1 Introduction

1.1 Background and Motivation

Climate change is already a reality. As global temperatures and sun exposure continue to rise, so do the cases of skin-related diseases. Among these, skin cancer is one of the most prevalent, with melanoma being one of the most common and concerning types.

Although melanoma ranks 14th among the most common cancers in Spain, some may perceive it as a lesser concern. However, its incidence rate per 100,000 inhabitants has been steadily increasing since 2003. The number of cases has risen from 12 to 15.1 per 100,000 in women and from 12 to 15.4 per 100,000 in men, highlighting a worrisome upward trend [1]. While previous studies have analyzed melanoma mortality rates across different provinces in Spain, none have examined its incidence from a spatial perspective.

1.2 Objectives of the Study

The primary objective of this study is to conduct a network analysis of healthcare facilities, specifically focusing on hospitals that treat cancer, in order to assess access to healthcare services. This analysis

will be cross-referenced with risk factors for melanoma and other relevant variables. In other words, the goal is to create a series of maps illustrating the melanoma situation across Spain's provinces, while considering all variables and factors that may influence this disease.

1.3 Risk Factors

- **Age:** Layer of mean age per province [2].
- **Gender:** Layer of men and women per province. Further analysis may be required, or it may not be included in the study [2].
- **Genetic Factors:** A risk factor, but it is difficult to represent in a spatial layer. Further analysis may be required, or it may not be included in the study [2].
- **Ethnicity and Skin Type:** Data has been extracted for this factor and classified using the Fitzpatrick scale (phototype of the skin) [3].
- **Number of Nevi:** A risk factor, but it is difficult to represent in a spatial layer. Further analysis may be required, or it may not be included in the study [2].
- **Sun Exposure and Geographic Location:** Layer of sun exposure hours per province [2].
- **Immunosuppression:** A risk factor, but it is difficult to represent in a spatial layer. Further analysis may be required, or it may not be included in the study [2].

2 Methodology

This study employs a comprehensive methodology combining data collection, cleaning, analysis, and visualization using various software tools. The approach follows a structured process, as outlined below:

2.1 Data Collection

- **INE:** National Statistics Institute of Spain. Collection of demographic and socioeconomic data, as well as immigration data [4].
- **AECC:** Spanish Association Against Cancer. Collection of melanoma incidence [5].
- **ArcGIS Online:** Collection of spatial data in the form of layers, such as hospital layers and the administrative boundaries of Spain.

2.2 Data Wrangling

- **R:** Cleaning and removal of missing values in some tables, as well as the creation of charts to visualize the evolution of data over time. Very helpful for dealing with the classification of skin phototypes.

- **Excel:** Creation of tables from web scraping and the creation of a common field to perform a join later on.

2.3 Data Analysis

- **ArcGIS Pro:** Once the data has been cleaned and processed, I am able to conduct spatial analysis by:
 - **Joining Fields:** Joining the tables to the layers by the Province Code.
 - **Calculating Percentages:** Creating a new field and using a query to filter.
 - **Filtering and Selecting:** By using queries, I can select, filter, and export the target data.
 - **Network Analysis:** Performing a network analysis on the layers of hospitals to analyze service facilities.
 - **Spatial Reference:** Changing the coordinate system for the area of Spain.
 - **Layouts:** Representing and visualizing the results through well-organized maps.

3 Case Study: Melanoma in Spain from a spatial perspective

With a population of approximately 49 million inhabitants, Spain is located in Western Europe [6]. It shares borders with France to the north and Portugal to the west, while the Mediterranean Sea and North Africa lie to the south. Administratively, Spain is divided into 17 autonomous communities—comparable to U.S. states—and 50 provinces—similar to counties in the U.S. Additionally, Spain maintains two small territories in North Africa: Ceuta and Melilla, often referred to as Spanish autonomous cities or protectorates (see Figure 1).

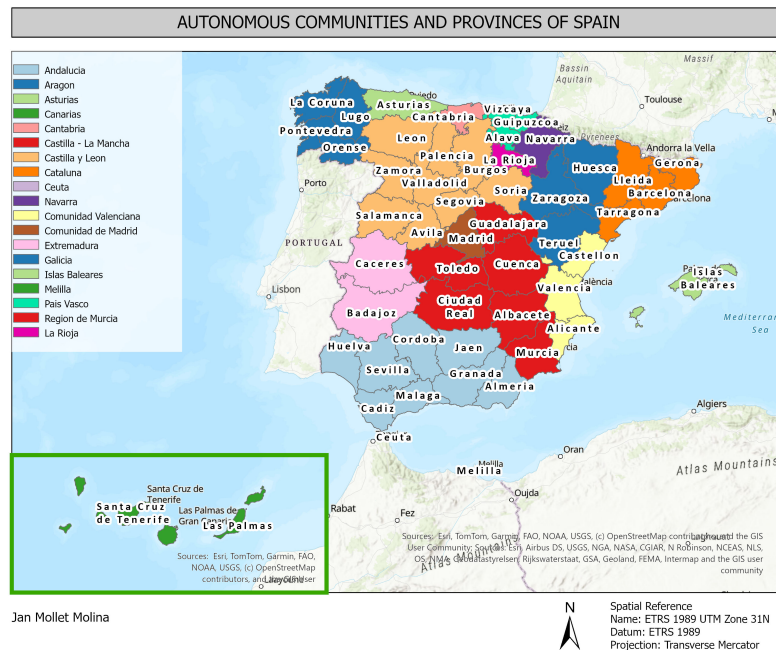


Figure 1: Autonomous Communities and Provinces of Spain

3.1 Population and Cases of Melanoma

Madrid and Barcelona are the two most highly populated cities in Spain, with approximately 3.5 million and 1.7 million inhabitants, respectively [6]. In the following map, melanoma cases are overlaid with the population distribution by province across the country. As expected, provinces with larger populations tend to report a higher number of melanoma cases. While this provides a solid starting point for the spatial analysis of melanoma, the map is not yet suitable for direct comparison between provinces, as it displays the absolute number of melanoma cases rather than cases relative to the population size in each province (see Figure 2).

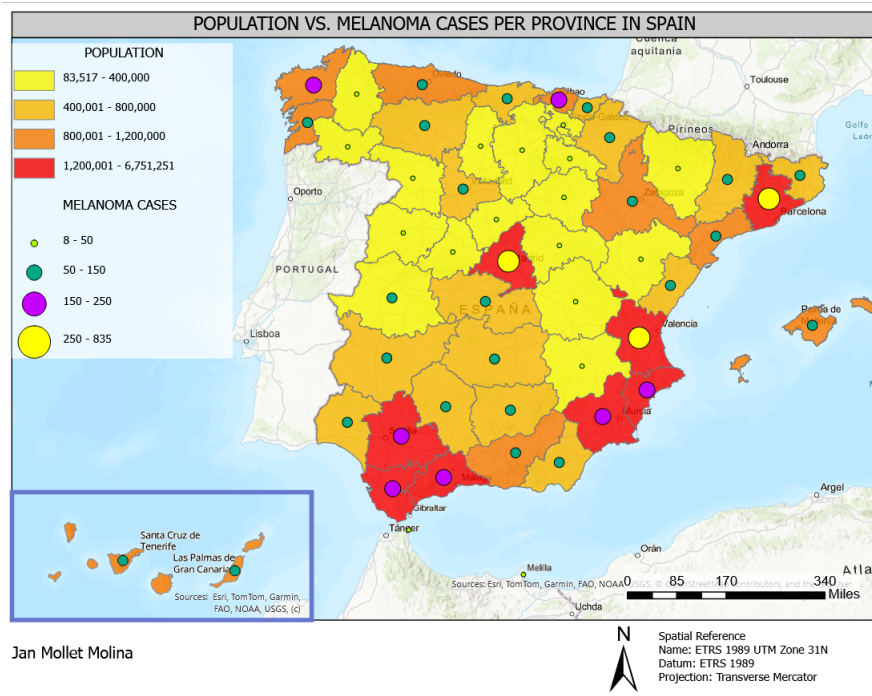


Figure 2: Melanoma cases across Spain (Absolute)

Given that, I have created a new field to calculate the number of melanoma cases per 100,000 inhabitants for each province in order to obtain relative results that allow for meaningful comparisons. To perform this calculation, I implemented the following formula:

$$\text{Melanoma Rate}_i = \frac{\text{Number of Cases}_i}{\text{Population}_i} \times 100,000 \quad (1)$$

Once the transformation was applied, I symbolized the layer using graduated colors based on the newly created field P_M_100K (melanoma cases per 100,000 inhabitants). While in the previous map I could have stated that the incidence of melanoma in Spain was higher in provinces such as Madrid, Barcelona, or Valencia, I can now confidently conclude that the highest incidence is located in the North-Western provinces of the territory (see Figure 3).

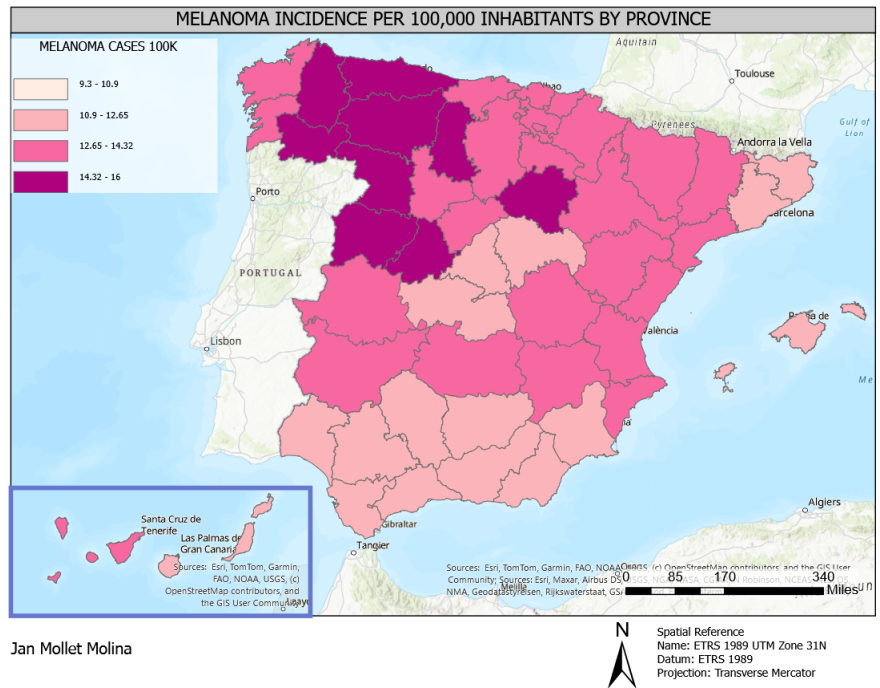


Figure 3: Melanoma cases across Spain (Relative)

3.2 The variables affecting melanoma

Now that we have a clearer understanding of the incidence of melanoma across the country, I will incorporate the variables previously discussed such as age, income, skin phototype, and sun exposure as well as access to hospitals (service facilities). These factors will help us detect spatial patterns and provide a more comprehensive picture of the melanoma landscape across Spain's provinces, considering the main variables that influence its incidence and geographic distribution. For the analysis, I will begin by representing each of these variables on individual maps to better understand their spatial distribution. Subsequently, I will conduct a network analysis and overlay the relevant layers to identify meaningful patterns and relationships that may provide deeper insights into the factors influencing melanoma incidence.

3.2.1 Age and Income

Age and income are two critical variables to consider in healthcare related analyses. Age is often associated with an individual's physical condition and their capacity to carry out daily activities or maintain a certain level of well-being. As people age, their vulnerability to various health conditions, including melanoma, tends to increase. However, income can still play an indirect role. Individuals with higher incomes may be able to afford private healthcare, which often provides faster access to specialists and preventive services. This can lead to earlier detection and treatment, ultimately improving health outcomes despite the universal public system. As shown in the map below, where bivariate colors are used to represent both Mean Age and Mean Annual Income simultaneously, the provinces in the north-west tend to have older populations and higher incomes. (see Figure 4) .

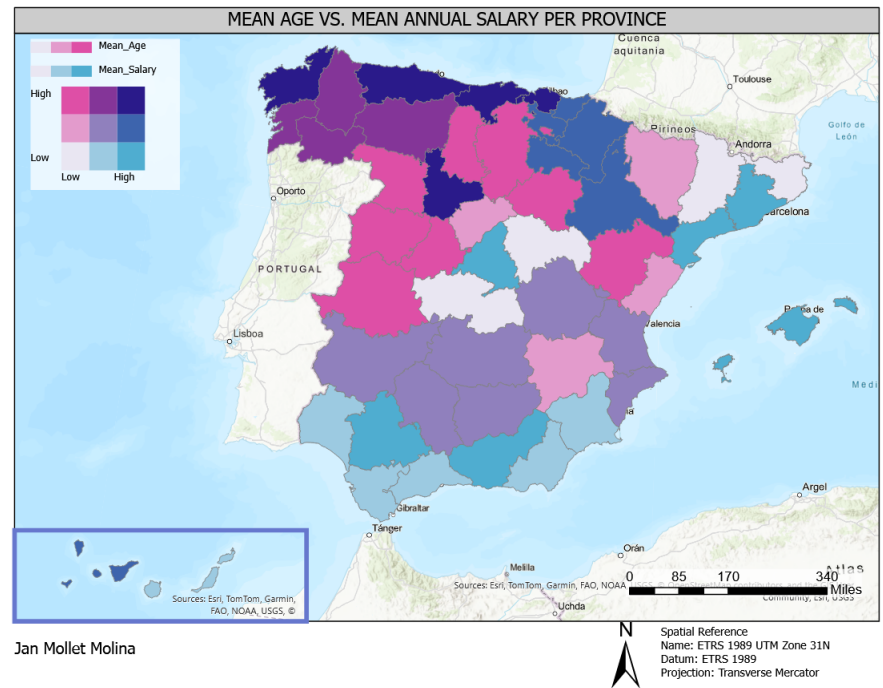


Figure 4: Mean Age and Mean Annual Salary in Spain

3.2.2 Skin Phototype (Fitzpatrick scale)

As explained earlier, one of the most important variables influencing the likelihood of developing melanoma is skin phototype. The Fitzpatrick scale is used to classify skin into six types, with Type I being the lightest skin tone and Type VI the darkest. To analyze this variable, I carried out a process to collect, store, and retrieve the relevant data, which was then imported into ArcGIS Pro as a CSV table (see Figure 5 and 6).

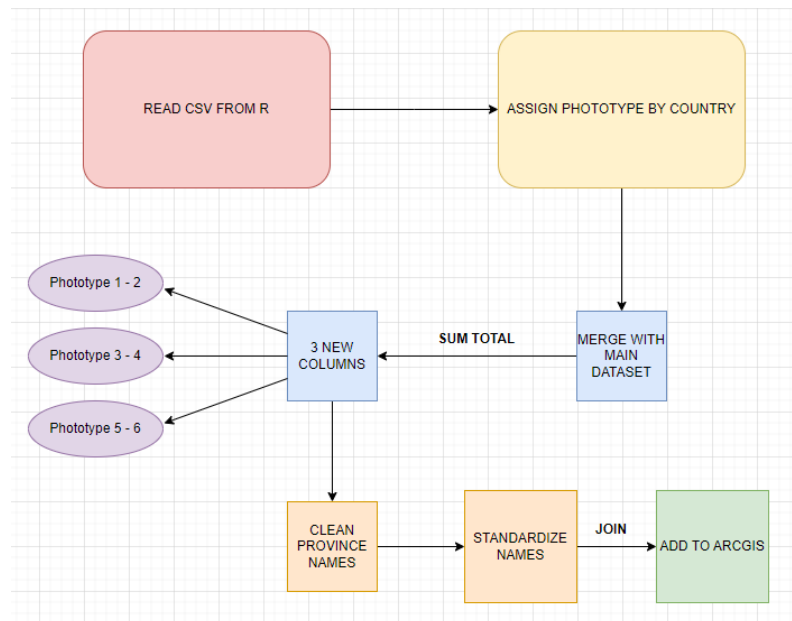


Figure 5: Data Processing Workflow for Skin Phototype Variable

To facilitate interpretation and improve clarity, I grouped the original six levels of the Fitzpatrick skin phototype scale into three broader categories: Types 1 – 2 (lighter skin tones), Types 3 – 4 (medium skin tones), and Types 5–6 (darker skin tones).

PROVINCE_NAME	Fototype_1_2_Total	Fototype_3_4_Total	Fototype_5_6_Total
ÁLAVA	1333	16016	14236
ALBACETE	1352	14951	9894
ALICANTE	76935	149841	76537
ALMERÍA	11046	47797	85000
ÁVILA	390	6171	3634
BADAJOS	708	17077	2980
ILLES BALEARS	35456	116874	45929
BARCELONA	77365	507155	259075
BURGOS	883	22555	5550
CÁCERES	612	6841	5944

Figure 6: R Output Table Showing Grouped Skin Phototypes

After joining the phototype CSV table to the provincial layer in ArcGIS Pro, I implemented the following formula to calculate the percentage of population of each skin phototype:

$$\text{Province percentage}_i = \frac{\text{Phototype}_{i,jg}}{\text{Population}_i} \times 100 \quad (2)$$

Once I had the three new columns with the percentages, I applied graduated colors to represent the three phototype categories (1 – 2, 3 – 4, and 5 – 6), allowing for clear visualization of their distribution across Spain (see Figure 7).

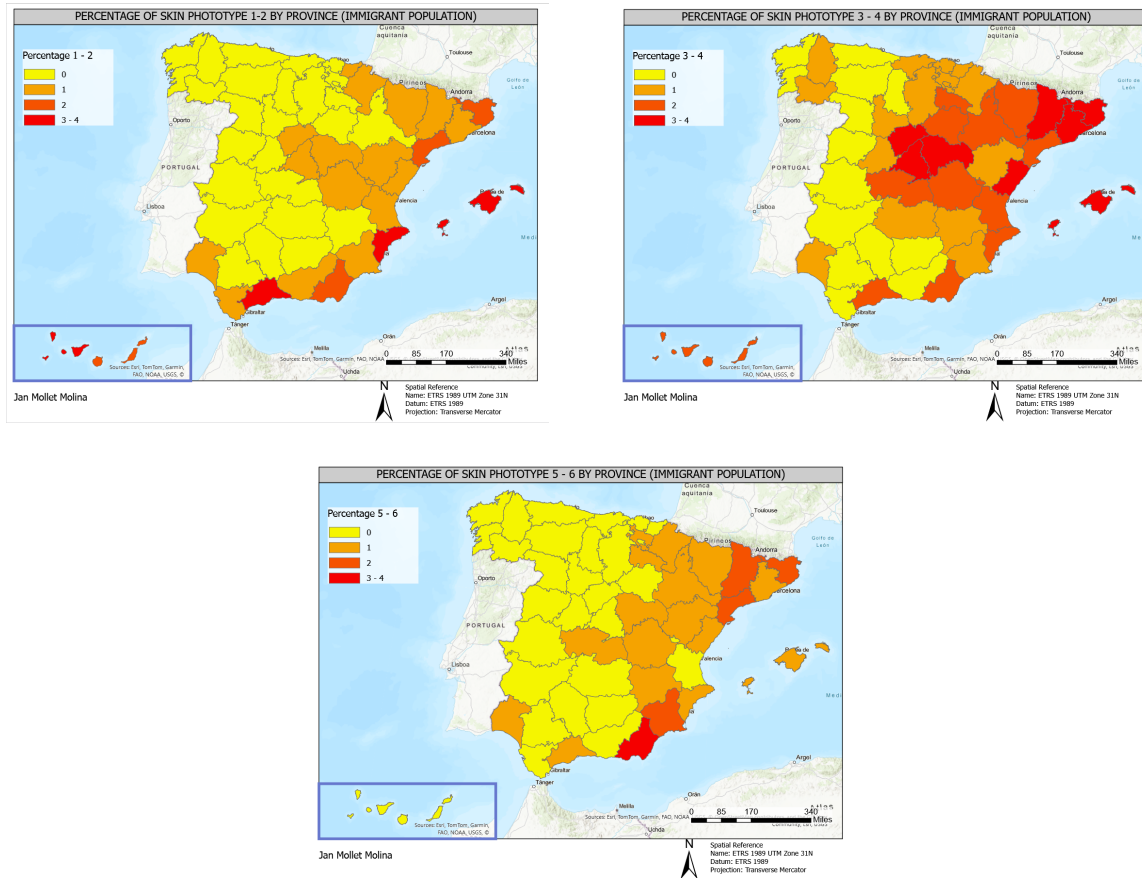


Figure 7: Maps illustrating the skin phototype (Fitzpatrick scale).

We have to take into account that I am only showing the percentage of the immigrant population by phototype, and not including people born in Spain (spaniards are consider to have Phototype II). So

they have a highly risk than someone from countries with phototype 5 – 6.

3.2.3 Hours of Sun Exposure

Now that I have mapped hours of sun exposure across Spain, I can clearly see how it varies from one province to another. Southern regions like Córdoba and Almería receive over 3,000 hours of sunlight per year, while northern provinces such as Cantabria and the Basque Country get much less. This variation in sun exposure is an important clue in understanding the geographic patterns of melanoma. The map helps highlight areas where people might have a higher chance of getting melanoma because they get more sunlight, giving us a strong starting point for analyzing how sunlight relates to skin cancer cases across the country (see Figure 8).

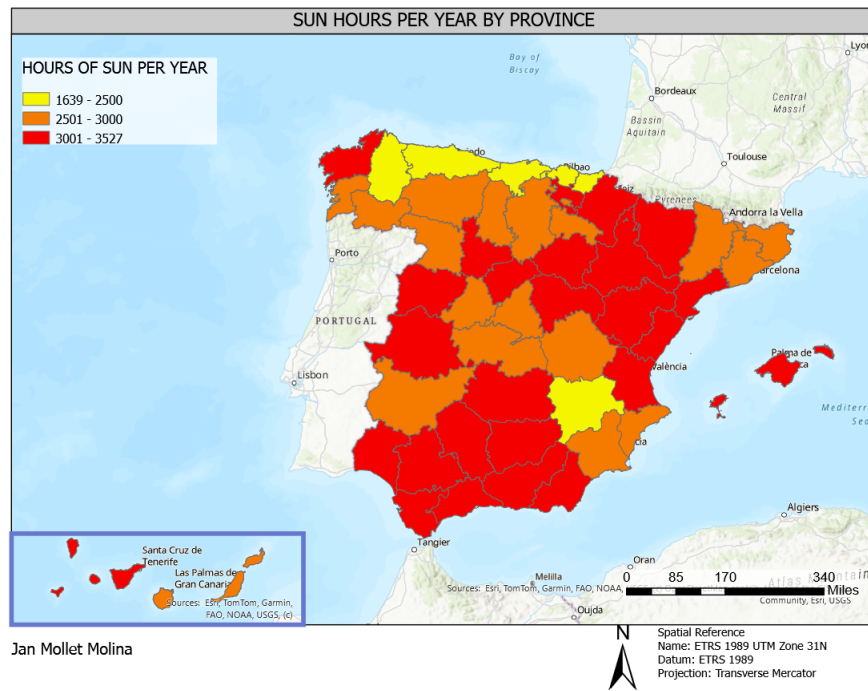


Figure 8: Hours of Sun per Year by Province

4 NETWORK ANALYSIS: Hospital Coverage

Now that we have a better understanding of how our variables are spread out across the country thanks to the maps and different types of symbolization I can move on to the main part of our project: the network analysis of service areas. In this case, I will focus on hospital coverage to see how easily people can reach healthcare services in different provinces.

4.1 Methodology for Network Area Calculation

I have followed different steps to calculate the area covered by the hospitals in each province.

4.1.1 Importing/ Filtering Hospitals

The first step was to import the hospital layers I downloaded from ArcGIS Pro Online into our project. I worked under the assumption that only large hospitals have oncology or dermatology departments, and therefore, they are the only ones equipped to treat melanoma. For this reason, I have filtered the hospitals by:

- **Number of beds in Hospital** field ≥ 100 beds
- **Type of Hospital** field == "General" OR "Oncológico"

With these criteria, I narrowed down our dataset, reducing the number of hospitals from approximately 900 to around 350.

4.1.2 Creating the Service Area Network Layer

Since I couldn't find a suitable road network dataset for Spain, I had to use the one integrated with ArcGIS Online by setting the different parameters:

- **Import Facilities:** Hospitals layers filtered
- **Cutoffs:** 30 minutes
- **Direction:** Away from facilities
- **Mode:** Driving time
- **Presicion:** Generalized
- **Option:** Dissolve

4.1.3 Obtaining the Area Covered by Hospitals per Province

Once the network analysis was completed and I obtained the polygons, I used the "Erase" tool to identify the uncovered areas within the province. Afterward, I calculated the geometries for both the uncovered areas and the province itself in square miles, creating two new fields for these values. I used the following formula to obtain the total covered area:

$$\text{Covered Area}_i = \text{Area Province}_i - \text{Area not covered}_i \quad (3)$$

Looking at the visualization:

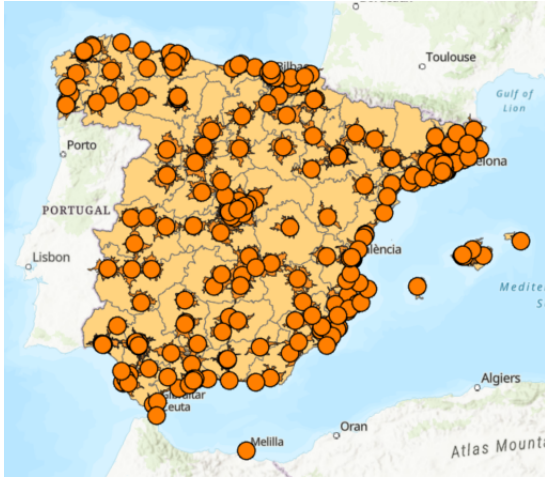


Figure 9: Screenshot showing hospital facilities.

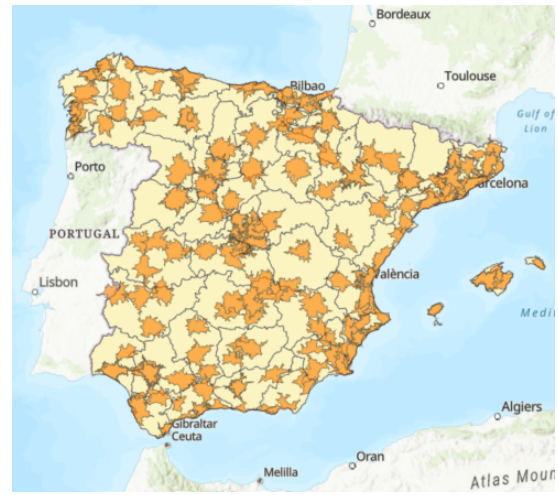


Figure 10: Screenshot illustrating the service area from the network analysis.

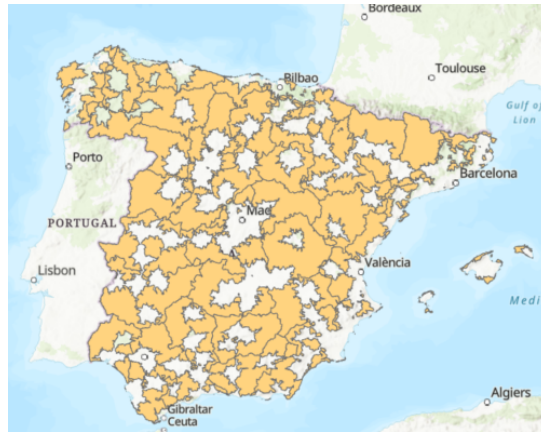


Figure 11: Screenshot showing the uncovered areas after using the "Erase" tool.

5 RESULTS AND CONCLUSIONS

5.1 Network Analysis

I was able to symbolize the new hospital coverage field along with the population for each province. I also included the join count of hospitals per province. Since I am representing three variables on the map, I chose to use bivariate colors, which make it easier to visualize relationships when dealing with more than two variables.

This first map provides a lot of information about the spatial distribution of hospitals and population coverage, while also showing the total number of hospitals per province.

A good way to interpret the map is by focusing on the colors at the corners of the legend grid (see

Figure 12):

- **Dark pink – top left corner:** Provinces with high population and low hospital coverage.
- **White – bottom left corner:** Provinces with low population and low hospital coverage.
- **Dark blue – top right corner:** Provinces with high population and high hospital coverage.
- **Sky blue – bottom right corner:** Provinces with low population and high hospital coverage.

In addition, the join count of hospitals provides further insight. Some provinces may have high hospital coverage despite having a relatively low total number of hospitals. A good example of this can be seen in the provinces of Murcia, Zaragoza, and Sevilla. These provinces are represented with a dark blue color, indicating high population and high hospital coverage. However, they are also marked with a green cross, which shows they have only 6 to 10 hospitals (general and oncology hospitals with more than 100 beds) a relatively low number considering their high coverage.

Based on these results, we can draw two possible conclusions:

- **Good distribution of hospitals across the territory:** Hospitals in these provinces may be strategically located to maximize accessibility to healthcare facilities, resulting in effective coverage despite fewer total hospitals.
- **Small area to cover:** The provinces might have smaller geographic areas, which could make coverage easier. However, we can confidently discard this explanation, as all three provinces, Murcia, Zaragoza, and Sevilla are among the largest in Spain in terms of surface area.

On the other hand, the provinces of Pontevedra and Vizcaya (the second smallest province in the country) show a high population and the lowest possible hospital coverage. Based on these results, we can draw two possible conclusions:

- **Poor distribution of hospitals across the territory:** Hospitals in these provinces may not be strategically located to ensure accessibility to healthcare facilities, which could explain the low coverage despite the high population. An additional factor worth exploring in future projects is whether these provinces experience particularly severe traffic congestion, which could make it more difficult for people to reach hospitals.
- **Larger area to cover:** It could be assumed that a larger geographic area makes healthcare coverage more difficult. However, we can confidently discard this explanation, as both Pontevedra and Vizcaya are among the smallest provinces in Spain in terms of surface area.

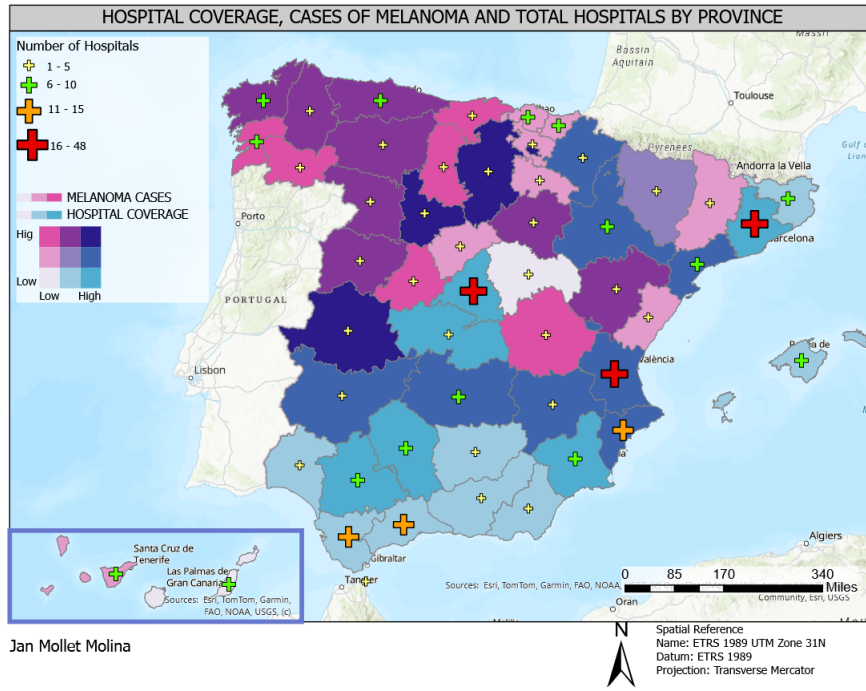


Figure 13: Crossing the Hospital Coverage and Melanoma Cases and also the Hospitals per Province

5.2 Weighted Vulnerability Index

The maps presented earlier provide valuable visual insights, clearly showing the distribution of hospitals, population, and melanoma cases across Spain. However, to gain a deeper understanding, I plan to go further by incorporating additional variables, such as Age, Income, Skin Phototype, Hours of Sun Exposure, hospital coverage, and melanoma incidence, as mentioned earlier.

How can these variables be combined to create a meaningful index that reflects the vulnerability of different regions in Spain?

5.2.1 Normalizing the Variables. Min-max scaling

Since all the variables were in the same table and layer, I was able to normalize them using the following formula:

$$\text{Normalize}_i = \frac{\text{Field}_i - \text{Min}_i}{\text{Max}_i - \text{Min}_i} \quad (4)$$

By applying this formula to each field, I obtained values between 0 and 1 for each variable, making them comparable.

5.2.2 Assigning weights for each variable

Once the variables were normalized, I created a new field and assigned a weight to each variable based on its relative importance. Here each assigned weight for each variable:

- **Cases of melanoma - Multiplier: 30%.**

- **Hospital Coverage - Multiplier: 25%.**
- **Hours of Sun Exposure - Multiplier: 20%.**
- **Skin Phototype - Multiplier: 10%.**
- **Age - Multiplier: 10%.**
- **Income - Multiplier: 5%.**

Using the following formula, we obtained the vulnerability index:

$$\begin{aligned}
 \text{Vulnerability Index}_i = & (\text{Melanoma}_{\text{Norm}} \times 0.30) \\
 & + (\text{Hospital Coverage}_{\text{Norm}} \times 0.25) \\
 & + (\text{Hours of Sun Exposure}_{\text{Norm}} \times 0.20) \\
 & + (\text{Skin Phototype}_{\text{Norm}} \times 0.10) \\
 & + (\text{Age}_{\text{Norm}} \times 0.10) \\
 & + (\text{Income}_{\text{Norm}} \times 0.05) = 1
 \end{aligned} \tag{5}$$

Now, these results have much more power and significance, providing stronger evidence for the relationship between the variables and enhancing the reliability of our conclusions. While the incidence of melanoma indicates the number of new cases, the vulnerability index can help predict areas at higher risk of mortality by considering key factors such as skin phototype, age, income, hours of sun exposure, and the existing incidence of melanoma (see Figure 14). These variables help identify populations that are at a higher risk of experiencing severe outcomes, including death.

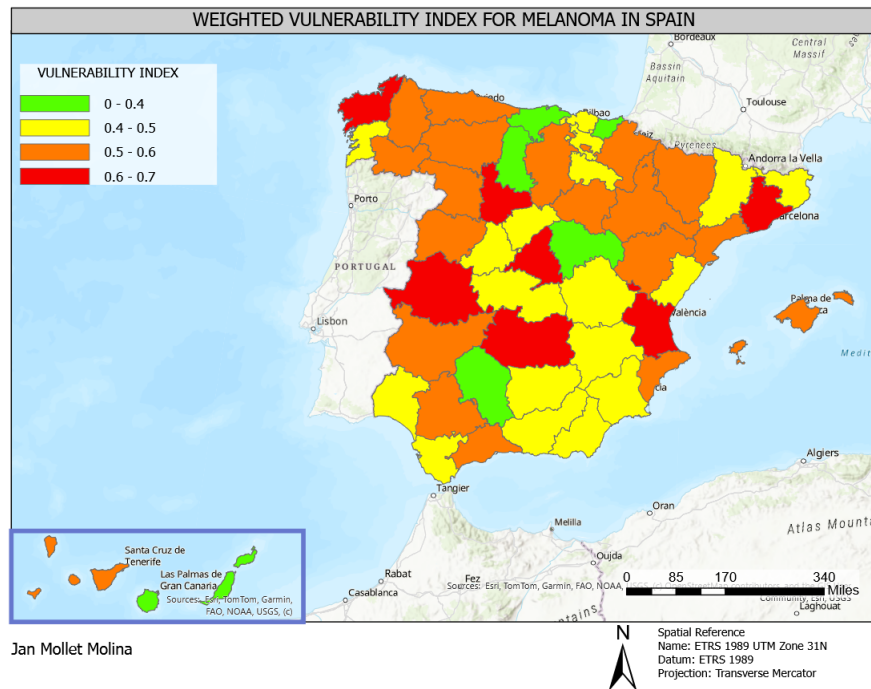


Figure 14: Vulnerability Index (REGRESSION MODEL)

Provinces shown in green are the least vulnerable to a higher risk of mortality, whereas those in red represent the most vulnerable areas that should be the focus of targeted prevention strategies.

6 FURTHER ANALYSIS

- **Municipality Data:** To improve our analysis, we could investigate municipality-level data, which would provide a more detailed understanding of how factors like population density, socioeconomic conditions, and access to healthcare contribute to melanoma risk. Examining data at this scale may reveal patterns and trends that are not apparent when considering larger geographic units.
- **Network Dataset (No Credits):** Another consideration is the inclusion of a network dataset, excluding credit-based variables. This would facilitate the examination of spatial relationships between regions, particularly in terms of hospital accessibility and transportation networks, which could impact the early detection and treatment of melanoma cases.
- **Machine Learning (KNN):** Using machine learning techniques such as k-Nearest Neighbors (KNN) could provide deeper insights into the factors influencing melanoma risk. By applying this algorithm, we can predict melanoma mortality based on various characteristics, helping to identify regions or demographics at higher risk according to historical trends.
- **Incorporating Additional Variables:** Future analysis could be enhanced by including additional factors such as lifestyle habits (e.g., diet and physical activity) and the incidence of melanoma by gender. These variables would provide a more comprehensive view of the factors influencing melanoma risk and how they interact.
- **Using the UV Index Instead of Hours of Sun Exposure:** Substituting the current measure of sun exposure hours with the UV index would enable a more accurate assessment of the link between ultraviolet radiation and melanoma risk. The UV index directly measures UV intensity, which could enhance the precision of our findings and offer better insights into how different levels of sun exposure affect melanoma development.
- **Melanoma Prevalence:** Adding melanoma prevalence would be ideal for understanding the total burden of melanoma in a population, considering both new and existing cases. By analyzing factors like incidence rates, healthcare access, and long-term care, we could assess how different regions and demographics are affected over time. This would offer a more comprehensive view of melanoma's impact and help guide prevention and treatment efforts.

References

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